

TANMAY SHARMA

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Education

Northeastern University

September 2024 – Present

Master of Science in Cybersecurity - 3.70/4.00

Boston, MA

Coursework: Network security, Linux administration, Cryptography, Binary exploitation, Linux rootkit development, ARMv8-A, x86, Security risk management and assessment

Technical Skills

Security Ops: Vulnerability Enumeration and Analysis, Secure Code Reviews, Active Directory, Linux Administration

Security Tools: Burp Suite, Metasploit Framework, Nessus, Wireshark, Nmap, Binary Ninja

Programming: C, Python, SQL, Bash, PowerShell, ROP, Socket Programming

Protocols & Standards: OWASP Top 10, CWE, MITRE ATT&CK, HTTP/HTTPS, SSL/TLS, OAuth, TCP/IP

Certifications: [CompTIA Security+](#), PNPT by TCM Security (In Progress)

Experience

Graduate Teaching Assistant

January 2026 – April 2026

Khoury School of Computer Science, NU

Boston, MA

- Conducting weekly lab sessions for a class of 59 students on network security, CTF Boxes, system security, Docker, and Linux administration topics etc while providing hands-on guidance and real-time debugging support.
- Assisting in development of supplementary course materials including lab documentation, Docker environment setup guides, etc while also grading hw assignments. Did QA and testing for Viceroy Decree's Sentinel CTF 2026.

Compliance and Security Consultant Intern

January 2026 – April 2026

Artelus.ai

Boston, MA

- Part of the penetration testing team, followed OWASP Web Security Testing Guidelines, performed reconnaissance, source code review, and exploitation across multiple sprints using Scrum methodology.
- Documented and triaged identified vulnerabilities with actionable remediation guidance, contributing findings to a broader regulatory evidence packet supporting U.S. and EU market entry.

Research Volunteer

March 2025 – August 2025

CyberspACe securiTy and forensIcs lab (CactiLab)

Boston, MA

- Collaborated on reverse engineering the EdgeTPU driver on Pixel devices, analyzing its interaction with firmware, which enhanced the lab's understanding of hardware security layers.
- Analyzed FirmXRay software by OSUSecLab, focusing on baseaddresssolver.java module, to identify how base addresses are resolved and features extracted, improving vulnerability assessment capabilities in embedded firmware.

Projects

Active Directory Lab | AD, LLMNR, Bloodhound, Kerberoasting

November 2025

- Designed and deployed a complete Active Directory pentesting environment with Windows Server 2022 Domain Controller and 2 Windows 10 Enterprise clients, configuring certificate services, domain forests, and realistic user scenarios to simulate enterprise infrastructure vulnerabilities
- Executed comprehensive security assessment covering 15+ attack vectors including LLMNR poisoning, SMB relay, Kerberoasting, and Golden Ticket attacks, documenting exploitation methodologies and defense strategies through systematic enumeration with BloodHound, ldapdomaindump, and PingCastle

Secure Instant Messaging System | Python, SRP, ECDH, TCP Sockets

January 2025 – April 2025

- Built a secure terminal-based chat system using SRP, ECDH, and AES-GCM for confidential, authenticated real-time communication (CY6740 - Network Security).
- Developed socket-based architecture with encrypted commands, rate limiting, and user listing, cutting abuse vectors by 80% and boosting resilience against attacks.

OSINT-Recon LLM Framework | Python, Mistral 7B, FAISS

October 2025 – Present

- Developed an AI-powered OSINT reconnaissance framework integrating Mistral 7B LLM with automated subdomain enumeration, DNS resolution, and HTTP service discovery tools for intelligent security analysis.
- Implemented RAG (Retrieval-Augmented Generation) system using FAISS vector database and sentence transformers for semantic search, with asynchronous tool orchestration generating automated threat intelligence reports.

Certifications, Achievements & Community

Achievements: [SimSpace Cyber Cup 2025 - Rank 6](#), Good rank in numerous CTFs

Community: Member of Boston Mesh Project and Boston Open Dev Community